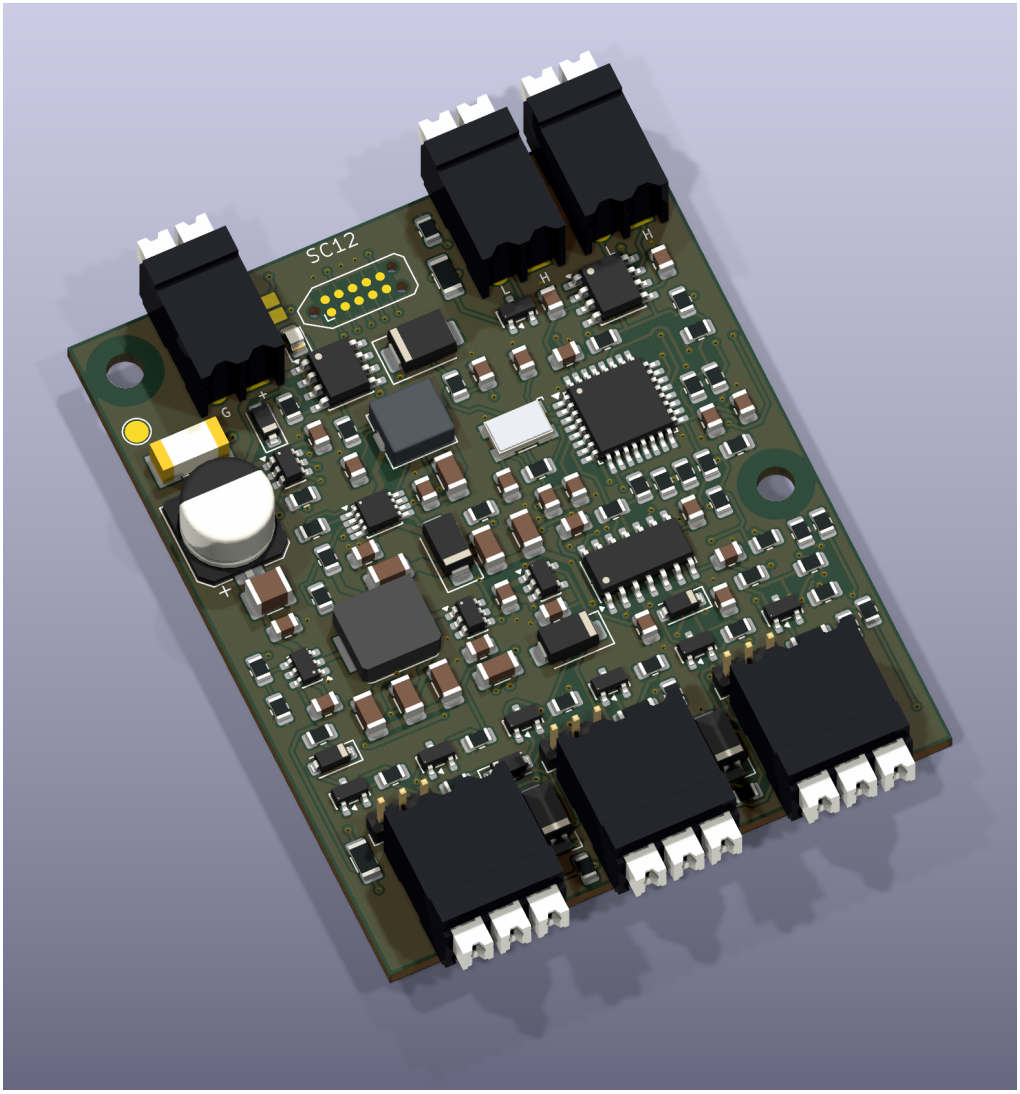
FRC
Sensor CAN

Revision F SC12 · Prototype engineering and assembly guide

Three industrial / logic sensor inputs
Regulated 12 V and 5 V sensor power



2026-09-08

Editable KiCad files, split factory/hand-solder BOMs, firmware, robot examples and verification records accompany this guide. No physical prototype has been tested. Electrical ratings are design targets pending the included qualification plan.

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1 Product overview

Source: README.md

Three digital sensor inputs over CAN, with regulated **5 V / 12 V sensor power** and software-selectable **NPN, PNP, push-pull, floating detection and keeper**. Removable 2 mm jumpers select each port's power voltage. All components mount on the top.

Unbuilt prototype. Engineering checks pass; startup, shorts, temperature, actual sensors, robot CAN and vibration still require physical qualification. This independent design uses its own protocol and drivers; it is not CTRE Phoenix or GrappleHook compatible.

Specification	Rev F
Board supply	6-22 V DC
Sensor supplies	Regulated 5 V / 12 V; **300 mA shared per rail**, not per port
Signal targets	Positive 0-30 V; LOW ≤1.2 V, HIGH ≥2.4 V; common ground
Input setup	Software bias and active-state selection independently per channel
Protection	Reverse input, fuse/TVS, separate latching current limits for 5 V sensors and the 12 V boost input
Fault reset	Remove the fault, then cycle board power; stored output-capacitor energy is not instantly removed
Controller	STM32F042K6T6, TCAN3413; 1 Mbit/s classic CAN
Software	App 0.7.0, bootloader 1.0.0, hardware 6, protocol 4
Servicing	CAN discovery, ID/settings and recoverable firmware upload; initial installation through SWD contacts
PCB / assembly	**52 × 64 mm**, four copper layers, 110 top SMT parts + nine hand-soldered parts + three loose caps

Rev F replaces E's battery-fed buck-boost stage with a protected boost from the existing internal 5 V supply. Cheaper equivalent capacitors and more compact placement lower parts purchasing from about **$35.14 to $27.17 per board at ten boards**, and PCB area falls **8.6%**. PCB, assembly, freight and tax are extra. Cost calculation and limitations.

Change voltage jumpers only with power disconnected. A missing cap disconnects port power, not its signal wire. Input mode changes use CAN software. Rail health is a coarse status indication, not calibrated voltage/current measurement. Analog readings and encoder decoding are not implemented.

2 Power and input circuit

Source: DESIGN_PLAN.md

The maintained native PCB and schematic implement the cost reductions below. The revision regression check compares the result with a frozen Rev E electrical/procurement contract. Historical full sources and packages remain available from the repository archive.

Supply path

```
J1: 6-22 V → fuse / reverse-polarity MOSFET / TVS → U1: internal 5 V, 2 A
                                                    ├─ U2: 3.3 V logic
                                                    ├─ U4: latch limiter → 5 V sensor rail
                                                    └─ U9: latch limiter → U8 boost → 12 V sensor rail
5 V / 12 V sensor rails → separate removable jumper on each port → sensor power
```

U1 remains LMR51420XDDCR. U8 is now TPS61085DGKR at 1.2 MHz, with the retained 3.3 µH shielded inductor and a B240A-13-F Schottky diode. Its nominal output is 12.003 V; the conservative static divider/reference budget is 11.64-12.38 V before transient and cable allowances. Three 22 µF output capacitors provide only about 10.8 µF after the modeled DC-bias/tolerance allowances. They must not be replaced by one nominally equivalent small capacitor.

A boost has a direct input-to-output path through its inductor and diode. U9 therefore sits **before the boost** and disconnects that path after an overload deglitch interval. It is the exact TPS2553DBVR-1 latch-off version, with a 21.5 kΩ limit resistor. Its roughly 1.2 A setting is an **input current limit**, not permission to draw 1.2 A at 12 V. The useful output target stays 300 mA. The 5 V sensor branch retains its own U4 limiter. Either latch requires a board power cycle to reset. Output capacitors still release stored energy into a short before it is interrupted; no reverse-powered output or independent output-overvoltage cutoff is supported.

At the modeled worst normal load, the boost needs about 1.02 A from the internal supply. The two maximum branch-limit budgets plus 100 mA logic total about 1.91 A, below U1's 2 A rating. Those calculations assume 80% boost efficiency and specified cable/regulator allowances. They do not establish thermal headroom or safe simultaneous fault transients on a physical PCB.

The unused fourth LM339LV comparator now monitors a divided 12 V rail. Its open-drain output and U9 fault share PG_12V. Nominal rising threshold is about 9.3 V, with a wide 8.0-10.6 V modeled range. It detects a missing/collapsed rail, **not** compliance with a sensor's minimum voltage. Firmware keeps the existing 100 ms health qualification. V12 startup, sensor readiness and thresholds need measurement.

U8 compensation is 13 kΩ / 33 nF. A 512-corner averaged CCM sensitivity screen passes its chosen margins, but its modulator-gain and amplifier bounds are engineering assumptions. It is not a vendor-validated switching model; DCM, cascaded-regulator interaction, startup, shorts and load steps remain bench tests.

Inputs, jumpers and layout

Each protected divider/comparator input and its software-controlled pull-up/down are retained. NPN, PNP, push-pull, floating and keeper modes remain independent across all three channels. Signal pull-ups use 3.3 V; selecting 12 V sensor power never changes them to 12 V. See input modes.

JP101/JP201/JP301 remain removable 2 mm power headers: 1-2 selects 5 V, 2-3 selects 12 V, and no cap turns off that port's supply. CAN end termination remains a separate solder bridge. There are no input-bias solder selectors or DIP switches.

The board is 52 × 64 mm, down from 52 × 70 mm. Four copper layers are retained: In1 is ground; In2 is ground with a short 12 V monitor branch and a 3.3 V service-contact connection. Converter switching/feedback/compensation and crystal nets stay on top without vias. All 119 fitted components are on top; the assembler places 110 SMT parts and the user fits six terminals and three headers. The user also supplies three removable caps.

Firmware identity is hardware 6 / application 0.7.0 / protocol 4, with separate image/configuration tags. Use this revision's complete package. Qualification and cost basis describe the remaining boundaries.

3 Wiring and programming contacts

Source: docs/WIRING.md

Use the included hardware-6/protocol-v4 firmware and tools; older-revision firmware must not be used on Rev F. Board coordinates below are viewed from the component side, origin at the top-left PCB corner. Bottom silkscreen is read after turning the board over. Do not infer pin numbering from left/right after flipping.

Connector	Pin 1	Pin 2	Pin 3
J1 power	Positive input, 6-22 V	GND	-
J2, J3 CAN	CAN H	CAN L	-
J4 input 1	Selected sensor power	GND	Signal
J5 input 2	Selected sensor power	GND	Signal
J6 input 3	Selected sensor power	GND	Signal

Each terminal pole has two solder legs; they are one electrical contact. J2/J3 are connected in parallel for bus continuation, with no onboard CAN ground terminal. Common ground comes through robot power. Use twisted H/L wiring; keep branches short. Strip 8 mm using the conductor range specified by Weidmüller for the exact terminal. Press the white release button for removal and for conductor types requiring it. Provide cable strain relief independently of the terminal.

Feed J1 from appropriately fused robot distribution. F1 is secondary board protection. Do not backfeed J4-J6 power pins, parallel the two sensor rails, or apply external power to the programming rail. The board is not galvanically isolated.

Sensor power and software bias

The factory leaves JP101/JP201/JP301 unpopulated for hand soldering. Fit the specified 2 mm headers, then leave caps off for initial bring-up. With power disconnected, fit ONE cap across 1-2 for 5 V or 2-3 for 12 V. No cap disconnects port power. Connecting all three power pins shorts the rails.

The headers are centered at X=7, 23 and 39 mm, Y=51.5 mm. Pins 1-2-3 run left to right from the component side; pin 1 is square. There are no physical input-bias selectors. Set each channel to NPN, PNP, push-pull, float detection or keeper in software; see input modes.

Use Harwin M22-2010305 headers and Sullins SPN02SVEN-RC closed-top caps (2 mm pitch; not 2.54 mm). Caps are friction-retained, not mechanically latched. After voltage/functional verification, support retention with a small removable bead of electronics-grade neutral-cure silicone between the cap exterior and header body, clear of the contacts. This is a proposed retention method; qualify it for your robot vibration/impact environment and enclosure. Allow full cure before use and inspect seating during service. Do not claim these are vibration-qualified merely because there is no slider. Bridge JP2 only at a CAN bus end that needs a 120 Ω termination. Leave it open at intermediate nodes; normally there are two terminations on the entire bus.

Examples: select `npn` for a sinking output (default active LOW), `pnp` for a sourcing output (active HIGH), or `push-pull` for an actively driven logic signal. NPN pull-up current is at most approximately 3.5 mA. Both NPN and PNP compatibility budgets allow 100 µA OFF leakage. No onboard 3.3 V sensor-power output is provided; the signal pull-up is not a sensor supply.

Design targets: 0-30 V positive input, LOW ≤1.2 V / HIGH ≥2.4 V, common-ground offset budget ±0.1 V. Do not use as an analog input, quadrature encoder, IO-Link port or continuous negative-voltage input. Sampling is 1 ms with 1-100 ms debounce; short pulses can be missed. Sensor power budgets are shared, 300 mA per rail. Both sensor branches latch off after a qualified overload; remove the fault and cycle board power to reset. The 12 V limiter is on the boost input, not an independent 300 mA output cutoff. Its rail monitor trips around 9.3 V nominal (wide tolerance), so a healthy flag does not prove a particular sensor has enough voltage. Rail-health indications are not calibrated voltage/current measurements. A fault on either rail makes the health-checked driver input unavailable. See protocol.

Representative industrial sensors

Regulated 12 V is a suitable nominal supply for the following families, subject to exact model and cable checks. Connect the selected output only; insulate unused complementary output wires separately. Brown/blue/black is common supply/ground/output coloring, but follow the specific datasheet rather than assuming colors.

Family	Published limits	Rev F setup / limitation
Omron E3Z three-wire NPN/PNP	12-24 V ±10%; typically 30 mA maximum, through-beam pair 35 mA; residual ≤1 V below 10 mA	Power 2-3 for 12 V. NPN or PNP software mode. Residual voltage fits the LOW target at this board's small load
Banner QS18 standard opposed/diffuse/retroreflective variants	10-30 V; <25 mA excluding output load; standard outputs <1 V at 10 mA	12 V supply is in range. Select NPN or PNP bias to match model; use one output. Account for its 100 ms power-up delay
Banner QS18 DVS/DL/FF variants	NPN residual specified up to 1.5 V at 10 mA	This exceeds the board's design LOW target. Prefer a compatible PNP variant or qualify actual NPN residual at the ~3.5 mA pullup load; do not assume all QS18 NPN models work

Sources: Omron E3Z specifications, Banner QS18 manual. PNP OFF leakage must also meet the input pulldown budget. Observe manufacturer cable/fusing requirements. Two-wire current-loop, AC, analog, IO-Link and safety/OSSD sensors are not supported as interchangeable three-wire inputs.

12 V is specified at the board, not guaranteed at the remote cable end under every transient. The analytical DC divider/reference budget is approximately 11.64-12.38 V before dynamic/distribution allowances; check ≥10.8 V at an E3Z during startup and worst battery/load conditions. Three 30 mA sensors are comfortably within the 300 mA shared design target, but include emitters, indicator loads, cable losses and inrush. The board's 100 ms rail qualification is not proof that every sensor has completed its own startup.

Programming contacts

J8 uses the **TC2050-IDC-NL mechanical pattern and a custom electrical mapping**, centered at (24, 5.75) mm. Use a mapped adapter, not an assumed ARM/STDC ribbon pinout. Ten 0.7874 mm contact pads are on 1.27 mm pitch, with three mechanical guide holes. In the component-side view, the upper row runs **10, 9, 8, 7, 6** left to right at Y=5.115 mm; the lower row runs **1, 2, 3, 4, 5** at Y=6.385 mm. Pin 1 is at X=21.46 mm, Y=6.385 mm. Use this U-shaped numbering, not an odd/even ribbon numbering assumption.

J8 pin	Signal	Connection
1	NRST	Probe reset, optional
2	MCU UART RX	3.3 V UART adapter TX
3	MCU UART TX	UART adapter RX
4	SWCLK	SWD clock
5	SWDIO	SWD data
6	+3V3	Probe VTref sense **only**, board output
7	GND	Probe ground
8	GND	UART ground
9	BOOT0	Normally leave open; onboard pulldown
10	GND	Ground

Power the board through J1 during programming. Leave adapter/probe power disconnected. BOOT0 is normally low. SWD programming and custom pogo-adapter fit require bench verification. The intended clip-free probe is held perpendicular by a fixture; the no-legs pattern does not retain itself.

4 Software input modes

Source: docs/INPUT_MODES.md

Rev F removes JP103/JP203/JP303. Each input has its own electronically controlled pull-up and pull-down. Select the mode in the desktop setup tool, CAN service CLI, or Java/C++ runtime API. This is an independent design with CANdi-like input choices; CTRE Phoenix and GrappleHook are not compatible protocols.

Mode in setup tool	Electrical behavior	Default active condition	Typical use
`npn`	Pull toward 3.3 V through a diode and 1 kΩ	LOW	NPN / open collector sinking output
`pnp`	Pull toward ground through 10 kΩ	HIGH	PNP sourcing output
`push-pull`	Both bias switches off	HIGH	Actively driven 3.3 V, 5 V or industrial logic
`float`	Alternate pull-up/down to classify the source	DRIVEN	Slow contact / high-impedance diagnostics
`keeper`	Bias toward the last sampled level; starts LOW	HIGH	Hold a released line near its last logic level

Active condition is independently selectable as `high`, `low`, `driven`, `floating`, `not-high`, or `not-low`. Inversion is applied afterward, then debounce. Normally leave the inversion mask at zero when using these active conditions. Firmware defaults to push-pull/HIGH on all channels; it does not guess a sensor type.

Wiring and electrical limits

Power, ground and signal remain the three sensor-terminal contacts. Choose 5 V or 12 V sensor power with the separate removable power cap, with board power disconnected. These power caps and CAN termination are not input-bias settings. Keep a common signal ground.

The pull-up uses the internal **3.3 V rail regardless of sensor supply**. Its series diode blocks most current from a 5-30 V signal toward the MCU rail. The voltage-rated pull-down MOSFET switches a 10 kΩ resistor. The LM339LV comparator, protected divider and hysteresis still receive the field signal; no field-voltage connection goes directly to an STM32 GPIO. This remains digital sensing, not analog acquisition.

Targets: positive 0-30 V input, LOW ≤1.2 V, HIGH ≥2.4 V, ±0.1 V ground-offset allowance. The compatibility budget allows 100 µA OFF leakage for NPN or PNP sensors. NPN LOW residual must remain ≤1.2 V at up to approximately 3.5 mA sink current. Verify the exact sensor datasheet. A nominal 5 V push-pull signal works with bias off. A 3.3 V open-drain output may use NPN mode if it tolerates the board's pull-up while the sensor is unpowered; otherwise use an external pull-up to its own supply with onboard bias off.

Both switches default off through external gate resistors during reset and in the bootloader. Firmware applies saved modes during startup and suppresses measurements until input/rail settling completes. Removing a power cap does not disconnect SIGNAL; an enabled pull-up can inject current into an unpowered sensor. Set push-pull/bias-off before servicing an externally connected unpowered device. Reverse diode leakage is finite: the 10 kΩ 3.3 V rail bleed budgets up to about 0.61 V unpowered rail rise using the BAT46W 60°C reverse-leakage specification. This is not isolation or a guarantee at higher temperatures.

Floating detection and counts

Float mode waits at least 2 ms after each bias change. Two matching full pull-up/down pairs are required, so initial classification needs approximately 8 ms, plus configured debounce for subsequent logical changes. Input-level telemetry reports HIGH, LOW, FLOATING or UNKNOWN (unsettled). Drivers mark logical inputs unavailable while any channel is unsettled or either rail is unhealthy.

Probe transitions never enter the logical edge counter. Counters track debounced inactive-to-active transitions. They pause during rail faults and maintenance; startup/reconfiguration re-baselines without inventing an edge. Use float mode for slow signals, and qualify the real cable capacitance, noise and sensor leakage. Fast changes during a probe pair can be misclassified; fixed NPN/PNP/push-pull modes are preferable

for ordinary sensing. This is not an encoder interface.

High impedance can mean a disconnected cable, an open contact, or the normal inactive state of an NPN/PNP output. Float detection cannot distinguish these causes. Keeper mode biases a released wire; it does not retain a value across reset or report a disconnection.

Save an example configuration

From `current/`, after discovery and physical identification:

```
python host/can_service.py --bridge localhost:5810 configure \
  --uid UID --device 17 --invert 0 --debounce 2 \
  --modes npn,pnp,push-pull
```

Replace `UID` with the discovered 24-digit hexadecimal serial. The tool stages all three settings, commits with the ID/settings record, resets, and verifies persistent readback. If `--modes` is omitted, existing saved input settings are preserved. Optional `--active low,high,high` overrides the defaults. The desktop window offers the same choices independently for each channel.

Java `configureInputsRuntime(InputMode[], ActiveWhen[])` and C++ `ConfigureInputsRuntime(...)` apply volatile settings. Read back `inputState()` / `ReadInputState()` to verify the configuration word and classification; runtime CAN commands have no acknowledgment and do not write flash. Use CAN service to persist. See wire format, service and drivers.

Engineering checks and simulations do not qualify electrical behavior. Bench-test all modes across supply, temperature, cable length, hot-plug, sensor startup and robot vibration before deployment. See validation.

5 Assembly

Source: docs/ASSEMBLY.md

Factory: fit only the 110 SMT references in Factory_BOM.csv using CPL_JLCPCB.csv. User: buy the nine through-hole parts and three closed-top caps.

Use the exact BOM, assembly drawing, interactive BOM and placement files. Reference labels are in these drawings; dense placement does not leave room to print every reference legibly on the actual top silkscreen.

1. Order the 52 ×64 mm, four-layer, 1.6 mm board with 1 oz finished copper target. Use the exported Gerbers and separate plated/nonplated drill files. Have the fabricator review its actual stackup, annular rings, mask registration and minimum features. Gold finish is useful at the programming contacts; keep solder and paste off them.

2. Check bare-board continuity and orientation. Apply paste to SMT pads only. A 0.10 mm stencil is a starting point; the assembler must review release and the VSSOP pad apertures. Through-hole header pads, solder selectors, TC2050 contacts and the test point deliberately have no paste.

3. Place 0805/1206/1210 passives, diodes and ICs. Verify C1 polarity, TVS/diode cathodes, pin 1 on each IC and the exact latch-off TPS2553DBVR-1 suffix. U8 TPS61085DGKR uses the supplied 0.65 mm-pitch VSSOP footprint. Confirm D4 cathode orientation and both U4/U9 TPS2553DBVR-1 latch-off variants. There are no components on the bottom.

4. Inspect joints and measure for supply shorts. Hand solder the six horizontal Weidmüller terminals after SMT reflow as the user assembly step. Then fit JP101/JP201/JP301: 2 mm headers on the top side, straight and fully seated, with pin 1 at the square pad. Use a removable cap temporarily as an alignment aid with the board unpowered; remove it afterward. Fill both plated holes per terminal pole. Support the board without loading fresh SMT joints. The terminal pins extend below the board.

5. Leave all selectors open for initial bring-up. Supply J1 from a current-limited bench supply, starting at 12 V and a modest current limit. Confirm input polarity, protected input, approximately 5.07 V internal, 3.3 V logic and 12.00 V regulated rails. Sensor headers remain unpowered while power selectors are open.

6. Program the combined factory image through the custom TC2050 mapping, set/verify boot protection, power cycle and configure a unique CAN ID using CAN service. Follow the fixture procedure. Save each input mode in software and verify readback. Then power off, choose each port's supply and termination settings, inspect one-cap-per-header placement, and check that 5 V and 12 V are not shorted.

7. Perform the qualification checks before connecting a robot harness. Begin with a known load and known logic source, then the actual sensor/cable.

Use M3 hardware at (3.5,4) and (48.5,36.5) mm. Holes are 3.2 mm NPTH. The copper keepout has at least 3.30 mm radius; use hardware whose contact area stays within a 6.4 mm diameter, with insulating washers/spacers as appropriate. Account for component bodies, connector overhang and protruding pins. Do not clamp the PCB between conductive surfaces. The mechanical drawing gives nominal geometry, not a fitted enclosure.

The compact board still benefits from magnification and a fine iron. The new power stage removes the former QFN and 0402 parts; the MCU and VSSOP still need controlled stencil/reflow assembly. See jumper retention and vibration qualification in WIRING.md.

6 Firmware

Source: firmware/README.md

Bare-metal STM32F042K6T6 firmware for three digital inputs. Build allocation is recorded by the current linker map/build report; stack depth has not been measured. The final two 1 KiB flash pages hold redundant 32-byte versioned configuration with CRC and commit-last writes.

Rev F uses **hardware identity 6** and **protocol 4**. It has a separate image tag and configuration magic so D2 images/settings are rejected. Use the matching drivers and software input modes. The separate power caps select sensor voltage.

CAN service is implemented, physically unqualified: full-UID discovery, identify LED, persistent ID/inversion/debounce and verified uploads. See CAN setup/update, robot bridge and qualification plan.

From `current/`:

```
source tools/env.sh
make -C firmware
bash scripts/test_firmware.sh
```

Outputs: `build/factory.hex` for initial SWD installation; `build/application.sc12` for CAN updates; separate bootloader/application ELF, BIN and MAP files. `sensor_can.hex`/ELF are application-only at 0x08002000. Flash: 8 KiB bootloader, 21 KiB app, 1 KiB manifest, 2 KiB configuration. Both linkers reserve the first 256 SRAM bytes and a 1 KiB minimum stack. The MCU controls six bias gates; no ADC sensing, sensor rail GPIO control or per-input LED logic is provided. The 8 MHz external crystal and PLL generate 48 MHz; bxCAN uses 16 time quanta with an 87.5% sample point at 1 Mbit/s. Auto bus-off recovery is enabled; automatic transmit retry is disabled.

Power through J1 from a bench supply. Use the custom TC2050 wiring: VTref sense→6, SWDIO→5, SWCLK→4, ground→7, optional reset→1. Leave probe power disconnected.

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -c 'adapter speed 1000' \
  -f tools/program_factory.tcl
```

This command has not been exercised on physical Rev F hardware. Verify MCU identity, supply and flash readback. Normal application programming should preserve the two configuration pages; full-chip erase removes them. Version-1/2 configuration is rejected.

UART: adapter TX→J8.2, RX→J8.3, ground→J8.8; 3.3 V, 115200 baud, 8N1; adapter power disconnected.

```
python host/sensor_can.py info --port /dev/ttyUSB0
python host/sensor_can.py configure --port /dev/ttyUSB0 \
  --device 17 --invert 1 --debounce 2
```

Device IDs 0-63 must be unique in this Team Use address space. Invert is a 3-bit mask; debounce is 1-100 ms. UART `SET id invert debounce_ms` and UID-targeted CAN service save persist and reset. Legacy CAN runtime configuration remains volatile. Unconfigured boards accept discovery/service but suppress normal telemetry. Power-cycle after initial SWD programming and verify boot protection before CAN updates. **Hardware sensor power is enabled independently of firmware**, including during reset or programming. On Rev F, leave the caps off JP101/JP201/JP301 to disconnect port power during initial bring-up; configure them only with input power disconnected. Input bias is electronic; CAN termination remains a solder setting. See assembly.

The independent watchdog is serviced only after the full 1 ms foreground cycle. A >20 ms foreground delay, clock failure, hard fault or PVD low-voltage detection puts CAN into standby; the watchdog provides reset recovery. The single LED pulses approximately 100 ms per second when configured and healthy, or 100 ms per 200 ms for configuration/clock/CAN errors. Rail faults also trigger the fast fault indication. PA3 reads 12 V PG and PA4 reads 5 V FAULT_N, both with internal pullups. A fault clears qualification immediately; recovery needs 100 ms continuously good indications. Counts freeze and debounce is re-baselined during this interval.

Broadcast Disable ID 0 stops telemetry and aborts pending transmissions. Local sampling and hardware sensor power continue. An explicit `RESUME` UART command or CAN resume command re-enables telemetry; reads never do this automatically. See protocol v4. Host tests cover firmware logic and mocked peripheral boundaries, not actual oscillator startup, electrical timing or STM32 peripheral behavior.

7 CAN protocol

Source: docs/CAN_PROTOCOL.md

1 Mbit/s classic CAN, extended 29-bit identifiers, 8-byte telemetry. This is an original Team Use protocol, not CTRE Phoenix. Device type=10 (Miscellaneous), manufacturer=8 (Team Use), unique device ID=0-63. Address = `(10 << 24) | (8 << 16) | (API << 6) | device`.

The APIs differ from A/B/B-2; Rev C shares these API numbers but uses protocol v2, which Rev F drivers reject. The version and capability checks prevent drivers from interpreting the wrong revision’s payload. Provision each board by full-UID CAN service or 3.3 V UART. An unconfigured board suppresses telemetry but answers service requests; its sensor power hardware is still enabled.

All multibyte fields are little endian. Masks use bits 0-2 for inputs 1-3. Unsupported versions, reserved bits, wrong lengths or stale frames must be reported as unavailable.

API	Period	Eight-byte payload
0x120 status	10 ms	version=4, sequence, debounced input mask, raw mask, channels=3, capabilities=0x7F, flags, reset flags
0x121 counts 1/2	100 ms	two uint32 rising-edge totals
0x122 count 3	100 ms	uint32 total, uint32 reserved=0
0x130 diagnostics	100 ms	uint32 transmit drops, TEC byte, REC byte, LEC byte, CAN flags byte
0x132 input state	Same as status	version=4, valid mask, floating mask, classified HIGH mask, uint32 input configuration
0x131 identity	1000 ms	4, firmware major=0, minor=7, patch=0, hardware=6, device ID, debounce ms, inversion mask

Status flags: configured=1, clock healthy=2, CAN initialized=4, bus-off=8, qualified 12 V power-good=16, qualified 5 V no-fault=32, all input classifications settled=64. Reset flags: power-on=1, pin reset=2, watchdog=4, software=8. Capabilities bits 0-4 indicate edge counts, runtime configuration, UART persistence, standalone hardware power and broadcast disable. Bit 5 indicates rail-health reporting capability; bit 6 indicates software bias/classification. Capabilities are static; flags bits 4/5 carry live qualified indications. The raw mask is the classified HIGH mask, excluding float probe transitions; check the valid mask in 0x132 before interpreting it. The debounced mask applies the active-state condition, inversion and debounce. Sequence is an unsigned byte and wraps. Counters wrap at 2^32 and clear on reset; runtime configuration preserves totals.

Diagnostic LEC is the STM32 three-bit last-error code (0-7); CAN flags are bus-off=1 and error-passive=2. There is no calibrated sensor voltage/current telemetry. The 12 V PG signal combines U9's active-low fault with a comparator threshold near 9.3 V nominal (modeled rising range 8.0-10.6 V); 5 V no-fault is an overcurrent-switch indication, not proof of voltage. Raw masks remain diagnostic data during faults.

A bad rail indication clears its flag at the next 1 ms sample; restoration needs 100 consecutive good samples. During faults and recovery, firmware re-baselines input debounce and freezes edge totals, preventing supply collapse/recovery from adding counts. All ports require both rail indications because sensor-power cap positions are not readable. Java/C++ input helpers also require fresh telemetry, configuration, clock health and CAN health and all classifications settled. They cannot detect every failed/disconnected sensor or open cable.

Command API	Payload	Effect
0x220 runtime input config	`53 43 04 invert debounce_lo debounce_hi 00 00`	invert 0-7; debounce 1-100 ms; volatile, does not change ID
0x222 runtime bias	`53 43 04 00` + uint32 configuration	Volatile, re-baselines input classification; read back 0x132
0x221 resume telemetry	`53 43 04 00 00 00 00 00`	Explicitly resume after broadcast disable

Extended arbitration ID **0** is the FRC broadcast disable. Firmware stops telemetry and aborts pending transmit mailboxes immediately on processing it. Sampling and standalone sensor power continue; there are no actuator outputs. Resume requires the explicit command above, UART `RESUME`, or reset. Drivers never resume from a read loop. Integrate explicit resume with a reviewed robot lifecycle policy; this implementation still requires robot-bus testing and does not establish competition approval. See WPILib CAN addressing.

Read timeouts: status 100 ms, counters/diagnostics 350 ms, identity 1500 ms. Counters 1/2 and 3 are separate snapshots. Sampling is 1 kHz; debounce requires a continuously stable logical state for the selected interval. This is not a high-speed pulse counter or encoder.

UART: 115200 baud, 8N1, 3.3 V logic. Commands `INFO`, `HELP`, `SET id invert debounce_ms`, `RESUME`. `SET` saves a version-4 redundant CRC-protected record and resets. The old four-argument command and earlier flash records are rejected; provision Rev F explicitly. No power-control command exists.

Service extension in firmware 0.7.0

Rev F retains protocol 4 with hardware identity 6; use only the matching drivers. APIs 0x300/0x301 at address 0 are reserved for CAN service. Opening maintenance pauses telemetry/counting on participating SC12 nodes; treat samples as unavailable immediately in robot code. Status period can be 10/20/50 ms until reset.

Per-input configuration word

Bits 0-2, 3-5 and 6-8 select channels 1-3 mode: 0=OFF, 1=PULL_HIGH, 2=PULL_LOW, 3=FLOAT_DETECT, 4=BUS_KEEPER. Bits 9-11, 12-14 and 15-17 select each active condition: 0=HIGH, 1=LOW, 2=DRIVEN, 3=FLOATING, 4=NOT_HIGH, 5=NOT_LOW. Bits 18-31 must be zero; other enum values are invalid. Default word zero means push-pull/HIGH on all channels.

Input state 0x132: a cleared valid bit means UNKNOWN regardless of the retained high/floating bit. For a valid channel, floating takes its own state, high means HIGH, otherwise LOW. High/floating masks never overlap. Validity/classification precedes active-state mapping, inversion and debounce. See mode timing and electrical limits.

At the default 10 ms period, telemetry is 231 frames/s; 20/50 ms give 131/71 frames/s. Status and input state are separate snapshots. Read timeouts are 100 ms for both; 350 ms for counts/diagnostics and 1500 ms for identity. Runtime configuration does not reset counters or persist to flash. Measurements settle after a mode change before further edges are counted.

8 CAN setup and updates

Source: docs/CAN_SERVICE.md

Rev F firmware **0.7.0**, bootloader **1.0.0**, service protocol **1**. Normal sensor messages remain protocol **4**, hardware **6**. Implemented and software-tested; actual STM32/robot-controller, power-cut and vibration qualification is pending. This is our own service protocol, not GrappleHook-compatible firmware.

First installation and recovery

Install factory.hex once through SWD using program_factory.tcl. This combines the protected bootloader, relocated application and image manifest. Rev F requires its own factory image and fresh configuration. D/D2 hardware cannot gain the new bias circuit through firmware. The script preserves the two configuration pages and refuses an unexpected device/flash size or an already protected bootloader. Its hardware execution is a pilot qualification gate.

From `current`, with J8 wired per the custom mapping and a separately powered, current-limited board:

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -f tools/program_factory.tcl
```

Power cycle afterward and verify `boot_protected: true` using the tool. Write protection covers flash pages 0-7 (OpenOCD protection blocks 0-1). Updating requires those blocks locked and application/configuration blocks writable. The script does not change readout protection or mass-erase configuration. Do not program `sensor_can.elf` or `.hex` alone as an initial install: these are relocated application-only files without the boot-valid manifest. Do not upload `factory.hex` over CAN.

Normal CAN updates use application.sc12. If interrupted, retry the complete file: there is no old-image rollback or cross-session resume. Invalid images remain in the bootloader. A normal boot gives a 500 ms discovery opportunity; an explicit boot request stays there, and three successive watchdog resets before two healthy application seconds also request recovery. `run` exits a valid bootloader image. SWD remains the recovery path for bootloader/option-byte damage or a nonfunctional CAN connection.

Laptop and robot use

Use Python 3.10+; the robot tunnel transport needs only the standard library. SocketCAN additionally needs `python-can`; configure the adapter for classical CAN at 1 Mbit/s. Tk is required only for the desktop window (`python3-tk` on many Linux distributions).

The robot bridge runs in a background thread alongside robot code. Copy `ServiceBridge.java` and the adapter for your WPILib year from examples/bridge. It listens only on `127.0.0.1:5810`; access it over an authenticated SSH tunnel. The 2026 adapter requires an explicit maintenance-request flag **and** Driver Station disabled state for every transmission. The 2027 alpha-7 adapter uses `RobotState` and an explicit `CANPort`, compiled separately with JDK 25. Choose a classical-CAN-compatible bus; this MCU cannot transmit CAN FD.

Keep the robot disabled, mechanically secure it and keep outputs inhibited in your robot code throughout maintenance. The bridge rejects or disconnects when permission/disabled state is lost; it cannot prevent someone enabling the robot at the Driver Station. No robot program is stopped or redeployed by this tool. Do not expose this unauthenticated TCP framing directly on a network. The sensor protocol itself has no authentication; CRC detects corruption, not a malicious uploader.

Example using a roboRIO reached over USB; use your controller's actual address/account:

```
ssh -N -L 5810:127.0.0.1:5810 lvuser@172.22.11.2
python host/can_service.py --bridge localhost:5810 discover
python host/service_gui.py
```

Copy the desired 24-hex-digit UID from discovery. Replace `UID` below with it:

```
python host/can_service.py --bridge localhost:5810 blink --uid UID
python host/can_service.py --bridge localhost:5810 configure \
  --uid UID --device 17 --invert 0 --debounce 2 --modes npn,pnp,push-pull
python host/can_service.py --bridge localhost:5810 update \
  --uid UID --file firmware/build/application.sc12
```

Direct SocketCAN uses `--channel can0`. Other commands: `info`, `run`, `rate --milliseconds 10|20|50`. Settings persist after save/reset; status period resets to 10 ms. Existing driver timeout is 100 ms. Traffic is 231/131/71

frames/s at 10/20/50 ms; verify bus jitter physically.

Discovery resolves the complete STM32 96-bit UID independently of assigned CAN IDs, including blank and duplicate-ID devices. Blink confirms physical selection. Before a change, the tool reads back all three UID words. ID conflict checks cover **responding SC12 devices only**; a powered-off board or another Team Use product can still have a conflicting address. Use one maintenance client per CAN network. The service APIs must be reserved against other Team Use devices.

Opening a session suppresses normal telemetry on all participating SC12 boards, and freezes/rebaselines input counts. Sensors remain powered by hardware. Drivers may retain their last sample until its normal freshness timeout; robot code must treat maintenance as unavailable immediately. Close/lease expiry restores normal sampling/telemetry unless Broadcast Disable previously latched it off. Broadcast Disable cancels maintenance and pending transmissions; explicit resume remains necessary for that latch. Never interpret a missing sample as a false input.

Wire protocol

Classical CAN, extended identifiers, eight data bytes, no RTR. Request ID `0x0A08C000`, response ID `0x0A08C040` (Team Use 8, miscellaneous type 10, APIs 0x300/0x301, address 0). Payload: opcode byte, sequence byte, nonzero little-endian session token (16 bits), little-endian value (32 bits). Response opcode OR 0x80 means success; OR 0xC0 means error. Echo sequence/token and validate the expected value. Errors 1-5 mean argument, state/offset, flash, image, and missing boot protection. No reply means an unmatched session/selection or transport failure.

Operation	Value / result
OPEN 0x01	Magic 0x314D4353; broadcast ACKs are identical; five-second lease
PREFIX0/1/2 0x02-04	Three raw UID words; no ACK; each change clears selection
PROBE 0x05	Prefix length 0-96, MSB first within each UID byte; matching nodes reply 1; only 96 selects
CLOSE 0x06	End session, no ACK
INFO 0x10	Index 0 flags/versions, 1 settings, 2 mode version, 3 upload offset, 4 capacity, 5 image validity, 6-8 UID words, 9 persistent input configuration
BLINK 0x11	Zero; three-second identify LED
SAVE 0x12	ID/invert/debounce/reserved bytes; reserved must be zero; explicit save then reset
BOOT / RUN 0x13/14	Zero; enter bootloader / run verified application
RATE 0x15	10, 20 or 50 ms; volatile status period
INPUTS 0x16	Stage validated per-input configuration; commit with SAVE; cleared on new selection/session
SIZE / CRC 0x20/21	Padded application length and CRC32; bootloader only
BEGIN 0x22	Product/layout tag 0x53433136; invalidate/read back manifest before application writes
OFFSET / DATA 0x23/24	Byte offset then four image bytes; ACK next offset after flash readback
FINISH 0x25	Zero; validate complete flash CRC/vectors, commit manifest last, ACK image CRC

INFO index 0 packs service version in bits 0-7, bootloader mode in bit 8, configured in bit 9, boot protection in bit 10, application minor 7 in bits 16-23, and hardware 6 in bits 24-31. Index 1 packs ID/invert/debounce/status-period as bytes; unconfigured ID is 255. Index 2 is major/minor/patch in bits 16/8/0. Index 5 returns 0 or 1.

Host discovery walks binary UID prefixes; matching responders send identical presence frames. Prefix frames are paced; UID readback guards selection before writes. Queries retry four times, probes twice; default pacing is at least 6 ms between requests and a 300 ms ACK timeout. Binary discovery can take seconds per device; intermittent connectivity can omit a node, so repeat discovery before commissioning. Uploads send OFFSET and DATA for each four-byte word, bounding throughput and working around flash stalls. Only the

immediately previous exact request is cached; offset/data logic also accepts an identical last word without rewriting it. Gaps, changed duplicate words and stale sessions are rejected.

Image and memory layout

Region	Address / size
Write-protected bootloader	0x08000000 / 8 KiB
Application	0x08002000 / 21 KiB
Image manifest	0x08007400 / 1 KiB
Configuration A/B	0x08007800 / 0x08007C00, 1 KiB each
Reserved SRAM	First 256 bytes: 192-byte vector table, validated recovery cookie

Both linkers reserve at least 1 KiB stack, excluding static data; runtime stack high-water measurement remains pending. The Cortex-M0 application copies vectors to SRAM and remaps them; bootloader handoff clears pending interrupts, stops SysTick/CAN and returns clocks to HSI. Compiled-register tests check vector copying, GPIO/CAN and configuration writes against peripheral models; they do not emulate analog effects or real interrupt timing.

The update file uses `<8s6I`: `SC12UPD1`, format 1, hardware 6, application address, padded length, CRC32, packed version, then image bytes. Length is 192-21504 bytes and a multiple of four. Initial SP must be aligned and within application SRAM, and Thumb reset PC inside application executable bounds. Manifest words are magic 0x49324353, size, CRC, product tag, CRC of the first four words, and final commit 0xC035A19E. Boot verifies manifest, vectors and full application CRC. No update command can write bootloader, configuration or option-byte regions.

Evidence and remaining gates

Tests execute the actual portable C engine with simulated flash, including 2,148 halfword erase/program power-cut positions on a two-page image, retries, corrupt files, wrong sizes, gaps, stale sessions, duplicate IDs and running-version verification. Compiled ARM tests exercise register intent and 34 configuration halfword interruption cases. Real TCP tests cover fragmentation and permission revocation with mock CAN. The Python uploader/GUI, bridge and firmware need end-to-end testing on real hardware before robot use. See qualification and unit programming/test fixture.

The `--modes` option and GUI persist independent input modes and active conditions using INFO 9 / INPUTS 0x16. Omitted modes preserve the existing persistent word; a runtime 0x222 change is not implicitly saved. SAVE commits ID, inversion, debounce and staged input settings together. See input modes.

9 Robot integration

Source: examples/README.md

Java 17 and C++ examples target WPILib 2026.2.1. Rev F uses protocol **4**, hardware **6** and firmware **0.7.0**. Use the matching `SensorCan12` and `SensorCan12Codec` files; older revisions are rejected. Provision a unique device ID through CAN service or UART.

Choose each sensor's 5 V / 12 V supply using the physical jumper before operation; the drivers do not select port voltage. See the Rev F setup guide and wiring.

CAN discovery, persistent settings and firmware updates are provided by the new service tools. The robot bridge has separately compiled 2026 and 2027 alpha-7 adapters. Sensor driver classes below remain pinned to 2026.2.1; the 2027 bridge does not port the normal drivers. Physical end-to-end CAN testing is pending.

```
private final SensorCan12 sensorBoard = new SensorCan12(17);
// In periodic(): absent means unavailable or a reported fault.
var detected = sensorBoard.input(1);
if (detected.isPresent()) {
  boolean sensorActive = detected.get();
  // Use a reviewed subsystem policy for this observation.
}
```

C++: `sensorcan12::SensorCan12 board{17};` then `board.ReadInput(1)` returns `std::optional<bool>`. Construct the driver once; an empty result must not silently become a valid “no object” observation. The helper requires fresh status, configured identity state, healthy clock/CAN and both qualified power indications. Raw `status()` / `ReadStatus()` remains available to diagnose faults. Counts are frozen during power faults and wrap at 2^32; compare/reset policy in the robot application.

Reads do not transmit or resume telemetry. `configureRuntime` / `ConfigureRuntime` sends volatile input inversion/debounce; it does not set persistent ID. `resumeTelemetry` / `ResumeTelemetry` is an explicit action after investigating broadcast disable, not a periodic automatic retry. Status timeout is 100 ms; count/diagnostic timeout 350 ms; identity timeout 1500 ms.

The Python host tool supports UART `info`, `configure`, `resume` and passive CAN `monitor`. UART uses 3.3 V levels at 115200 8N1; disconnect adapter power. CAN monitor requires a suitable external CAN interface; a UART adapter is not a CAN interface. Run the following from `current/`:

```
source tools/env.sh
python host/sensor_can.py configure --port /dev/ttyUSB0 --device 17 --invert 1 --debounce 2
python host/sensor_can.py monitor --interface socketcan --channel can0 --device 17
bash scripts/test_drivers.sh
```

Tests cover codec boundaries, all reserved status bits, stale/malformed responses, fault rejection, unsigned counts and explicit-only writes. The C++ example compiles against the actual pinned WPILib headers; Java compiles against pinned WPILib jars. Hardware bus behavior and competition acceptance remain open qualification steps. See protocol, wiring, and validation.

Independent input modes

Use the desktop service tool to persist per-input settings. Runtime Java example (volatile; verify `inputState()` readback):

```
sensor.configureInputsRuntime(
    new SensorCan12Codec.InputMode[] {
        SensorCan12Codec.InputMode.PULL_HIGH,
        SensorCan12Codec.InputMode.PULL_LOW,
        SensorCan12Codec.InputMode.OFF},
    new SensorCan12Codec.ActiveWhen[] {
        SensorCan12Codec.ActiveWhen.LOW,
        SensorCan12Codec.ActiveWhen.HIGH,
        SensorCan12Codec.ActiveWhen.HIGH});
```

C++ exposes `ConfigureInputsRuntime` with `std::array<InputMode,3>` and `std::array<ActiveWhen,3>`, and `ReadInputState()` for classification/configuration. `input()` / `ReadInput()` require fresh, healthy status and all inputs settled. Level helpers return UNKNOWN when the channel valid bit is clear. See mode behavior and electrical limits.

10 Purchasing and factory quote

Source: docs/FACTORY_ORDER.md

Use RevF_SC12_Factory_Upload.zip for a **prototype lot**, after reviewing the validation plan. No board has been physically qualified, no factory quotation is included and no order has been placed. The exact-MPN BOM is intended for a Chinese or other assembler that accepts customer-specified sourcing; it is not a pre-approved JLCPCB/LCSC turnkey basket.

For a JLCPCB upload, start with the fabrication-only Gerber ZIP, then supply BOM_JLCPCB.csv and CPL_JLCPCB.csv. Keep Factory_BOM.csv alongside for exact-MPN matching. The larger factory packet is a complete reference handoff.

Fabrication

Setting	Requirement
Board	52 × 64 mm, four copper layers, FR-4, 1.6 mm nominal
Layer order	F.Cu signal/power, In1.Cu GND, In2.Cu GND + 12 V monitor and 3.3 V service branches, B.Cu signal/power
Copper	35 µm nominal each layer; verify finished copper with supplier
Dielectric	Symmetric four-layer stack; target outer-to-inner dielectric ≤0.25 mm; supplier must confirm final stack
Finish	ENIG specified for exposed programming contacts and fine-pitch ICs
Minimum trace	0.18 mm
Clearance	0.20 mm general; U8 package pad-to-pad 0.15 mm local minimum
Vias	Through vias, minimum 0.30 mm drill / 0.60 mm pad; no blind/buried vias
Drills	Separate PTH and NPTH Excellon; no plated mounting or alignment holes
Solder mask	Both sides; preserve fine-pitch openings; review U8 mask webs with CAM
Stencil	Top only, nominal 0.10 mm laser-cut; use supplied paste apertures
Panels	Factory rails/fiducials may be added outside the delivered outline; preserve connector overhang

Native nominal stack: 0.035 mm copper on all four layers, 0.20 / 1.04 / 0.20 mm dielectric and 0.01 mm solder mask per side, totaling 1.60 mm. These are planning thicknesses; final fabrication tolerances and material selection belong to the approved supplier stack.

This stack is an order requirement, not a confirmed laminate/impedance quotation. CAN is short board-level routing; no fabricated impedance coupon has been qualified. Get the supplier's CAM/stackup review before fabrication.

Assembly

Factory scope: 110 SMT components on the top side only. Use `bom/Factory_BOM.csv` (exact MPNs) and `output/assembly/CPL_JLCPCB.csv`. The simplified `bom/BOM_JLCPCB.csv` is a matching aid; blank LCSC numbers require exact-MPN matching against Factory_BOM, never an automatic substitution.

User scope: six terminal blocks plus three voltage headers (nine through-hole parts), then three loose caps. The factory must leave J1-J6 and JP101/JP201/JP301 empty. `positions_all_top.csv`, the complete BOM and drawings describe the finished board, not additional factory placements.

CPL coordinates are mm from the PCB lower-left, positive X right and Y up, with native KiCad counterclockwise rotation. Match the uploaded preview to the assembly drawing and check IC/diode pin 1; machine-library rotation conventions can differ.

Power headers are unpopulated; the JP2 termination bridge and all J8 contacts are **unbridged/unpasted as supplied**. The factory must not add solder across JP2 unless a separate reviewed port-configuration work instruction is supplied. The user will hand solder all through-hole parts after delivery. The family name “LSF-SMT” does not make these exact connectors ordinary two-pad SMT parts.

Use exact TPS61085DGKR for U8 and latch-off TPS2553DBVR-1 for both U4/U9. Do not substitute auto-retry TPS2553, a lower-saturation inductor, or nominally equal small MLCCs without electrical review. Preserve capacitor voltage/dielectric/DC-bias behavior, the precision resistors and the 13 kΩ / 33 nF compensation network.

Inspect VSSOP/MCU lead joints, diode polarity and all passive values. Perform unpowered shorts/selector checks, staged current-limited bring-up, firmware programming/readback and rail/input/CAN tests. Track every component substitution and lot.

Sourcing, cost and delivery

Cost reduction and batch estimates separates parts procurement from assembly. LCSC inventory is not proof of JLCPCB factory allocation. No SMT part allocation, complete cart or final quotation has been confirmed. Request exact-MPN matching and incoming date confirmation before paying; the STM32 and any unavailable exact MPN may require outside procurement. Any unapproved item must remain unresolved rather than being silently substituted.

Request Economic PCBA first if the exact parts and four-layer board qualify; compare Standard PCBA with the same SMT-only scope. Avoid paying for bottom-side assembly or through-hole work. Standard assembly still charges feeder loading for all 48 SMT types. No programming or functional-test service is included by this package unless separately quoted; the user performs bring-up and programming.

One-month delivery is unconfirmed. Obtain allocated-part dates, CAM/assembly timing and a destination-specific tracked express quote before scheduling the robot build. Procurement holds and customs may exceed the target. Order hand-fitted parts in parallel once the BOM is finalized.

Programming deliverables

The packet includes combined `firmware/build/factory.hex` and `tools/program_factory.tcl` for initial SWD programming/protection. Power cycle and verify protection, then use CAN for setup and later `application.sc12` updates. User programming is the default quoted scope; see fixture procedure. Do not substitute an application-only ELF/HEX.

11 Qualification

Source: docs/VALIDATION.md

No Rev F prototype has been assembled, powered or tested. The files support engineering review and prototype fabrication. The production/competition qualification gates below remain open, even when every automated check passes.

Automated evidence

The release contains native KiCad electrical/design-rule reports, explicit pin-model/schematic/PCB parity, placement/selector/programming-contact checks, manufacturing-file parsing and copper checks, host firmware tests, and Java/C++/Python protocol tests. These check design consistency and modeled behavior. Mocked peripheral tests do not prove STM32 timing or CAN electrical operation.

Electrical checks sweep input-threshold tolerances, divider accuracy, resistor stress, fuse/load budgets and manufacturer-curve capacitor derating. The separate converter model screens boost CCM loop corners only. Neither substitutes for hardware measurement. Reports state their assumptions rather than treating typical values as guaranteed bounds.

Prototype gates before production release

Gate	Measurement / acceptance intent
Assembly	Confirm exact horizontal terminals, pin-to-hole fit, polarity, U8 VSSOP joints, D4 polarity, U9 latch-off suffix, selector isolation and no solder shorts
Staged power	No sensors first; current-limited 12 V source, verify 5 V/3.3 V before full operation, then regulated 12 V and low-load input current
Regulation	Sweep 6-22 V with 0-300 mA on each rail separately and together; verify 12 V remains within the selected sensors' supply limits at their cable ends
Cascaded startup	Scope V5_INTERNAL, V5_BOOST, BOOST_SS, SENSOR_12V and 3V3 during slow/fast input ramps, repeated starts and brownouts. Confirm boost startup/inrush does not falsely latch U9 or collapse U1; review operation below 6 V
Transients	6↔14↔22 V line steps; 0↔300 mA load steps; representative sensor startup/inrush and cable lengths; check ringing/undershoot on both cascaded converters
Stability	Measure loop response or justified equivalent transient qualification on the cascaded 5 V buck / 12 V boost, including CCM and light-load/DCM modes; target >45° phase and >10 dB gain margin where measured
Faults	Short each sensor supply and each port; verify current-limit behavior, both latching branches and power-cycle reset, firmware flags, no spurious counts and safe temperatures. Measure peak current and output-capacitor discharge energy before U9 disconnects; verify cold-start into a short and hot-plug loads
Electrical protection	Reverse input, realistic protected-branch fault energy, fuse clearing/I^2t, inductive transients and TVS stress using suitable test equipment. Not automotive load-dump qualified
Temperature	Both rails fully loaded at 6/12/22 V, intended ambient and later enclosure; measure U8/U1/L1/L2/Q1/fuse/terminal and MCU temperatures, hot-start/restart and surface safety
Inputs	0-30 V PNP/NPN/push-pull thresholds, off-state leakage/residual voltage, ground offset, sensor disconnection, wiring transients and all intended sensor models
CAN/firmware	Real roboRIO bus at 1 Mbit/s with correct end termination; no normal telemetry before provisioning, unique IDs, bus-off, clock faults, watchdog, broadcast disable, explicit resume, stale data and power-fault handling
EMC/mechanical	Motor-noise environment, ESD at terminals, cable retention, vibration, mounting insulation and programming-adapter fit; repeat after enclosure design
Production process	Approved stackup/CAM, stencil/profile, supplier alternates, test fixture, serial/lot records, programming/provisioning and final functional limits

FRC integration

Use a single appropriate protected power-distribution branch and wiring sized for its protection. The 0451 3 A fuse has a published 300 A interrupt rating at 32 V DC; confirm prospective branch fault current and clearing energy for the actual harness. The onboard fuse is secondary protection, not permission to bypass the robot distribution rules. Sensor grounds are common with the board; isolate conductive sensor cases where required by frame-ground rules. Do not use dedicated radio/roboRIO power outputs as a convenient supply.

The custom Team Use CAN address uses manufacturer 8 and miscellaneous device type 10. The implementation processes broadcast disable ID 0; it has no actuator outputs. It is not a CTRE Phoenix device. Robot inspection and the current season's rules still apply. Commercial CAN product acceptance, assigned identifiers and required ecosystem support are separate outstanding steps; Java/C++ examples alone do not establish approval.

Primary references: 2026 FIRST manual, especially custom-circuit voltage, distribution/wire and frame-ground rules; WPILib CAN addressing and acceptance requirements. Recheck applicable rules for the actual event/season.

Rev F-specific prototype checks

The 12 V monitor is a coarse collapse detector (~9.3 V nominal, modeled 8.0-10.6 V rising threshold), not a precision undervoltage guarantee. Measure actual sensor-terminal voltage and startup separately. The boost input limiter is approximately 1.2 A; the useful 12 V output remains 300 mA. Check 5 V source current and temperature with both rails loaded, and verify that a fault on either branch does not cause unsafe shared-supply oscillation. The 512-corner CCM result uses assumed amplifier/modulator bounds and does not qualify silicon behavior.

Verify the exact 2 mm header and cap fit, both 5 V / 12 V positions and unpowered open state on every channel. Verify no connection between the two rails in either position. Check solder fill on all 39 user-fitted through-hole joints. Fit/retention, vibration and impact testing must include the actual caps, chosen removable retention compound, enclosure and robot harness; these headers have no positive latch. Compare the substituted passive values/ratings against the supplied BOM. Factory BOM and CPL must contain exactly 110 SMT references, with all nine through-hole components omitted. Firmware identity is hardware 6 / protocol 4; prior-revision firmware is rejected.

CAN servicing qualification

The service engine and host uploader pass simulated flash cuts/retries; ARM register checks and both bridge adapters compile. Before robot use, measure actual boot/vector/interrupt behavior, option-byte protection, stack use, upload time, CAN load, bus-off recovery and duplicate-ID discovery. Interrupt power during invalidation, every erase/program phase and final commit on physical units. Verify persistent configuration, bootloader recovery, controller disconnect/enable rejection and healthy telemetry after exit. Protocol and evidence; per-unit fixture records. No software simulation closes these physical gates.

Electronic bias qualification

Verify Q101/201/301 PMOS and Q102/202/302 NMOS polarity and correct exact MPNs. Scope all six gate signals during reset, SWD programming, bootloader, watchdog recovery and supply ramps: pull-up gates HIGH and pull-down gates LOW mean both biases off. Verify mixed NPN/PNP/push-pull operation and nonvolatile readback independently on all three channels.

Sweep external signal 0-30 V in every mode. Measure inactive sensor leakage margins, active residual voltage and pull-up sink current (~3.5 mA maximum). Test 3.3 V and 5 V push-pull, unpowered sensors and unpowered board backfeed. Qualify the BAT46W leakage/rail-bleed budget over the actual operating temperature; current calculation uses its 60°C reverse-leakage specification, not a zero-backfeed claim.

Test floating detection with open contacts, NPN and PNP inactive outputs, long cables and added capacitance. Verify UNKNOWN during inconsistent probe pairs, no probe-generated edge counts, and one debounced edge on actual active-state transitions. Check keeper retention while released and LOW initialization after reset. Verify fixed input modes for faster signals; float detection can alias changing sources.

12 Unit fixture

Source: docs/FIXTURE.md

Build a reusable fixture around the documented custom TC2050 contacts, ST-Link, a separately current-limited J1 supply, accessible rail test contacts, CAN adapter and three independently driven input fixtures. This document and scripts are ready for the pilot; no physical fixture or unit test has been performed.

Leave voltage caps absent for initial programming. Keep the SWD probe's voltage pin as target reference only. Program/verify the combined image and set boot write protection with program_factory.tcl, then power cycle and verify protection over CAN. Save the programmer log, factory image SHA-256, component lot and full MCU UID. The setup checks STM32F04x device ID 0x445, 32 KiB flash and WRP pages 0-7 (OpenOCD protection blocks 0-1, not 0-7), consistent with the OpenOCD STM32 driver. Qualify the exact probe/OpenOCD version on the first unit.

Configure a unique ID through the service tool. Record internal 5 V, logic 3.3 V and regulated 12 V with a calibrated meter; acceptance windows are 4.8-5.3, 3.15-3.45 and 11.4-12.6 V. These generous unit-screening windows do not replace rail regulation/load/temperature qualification. Power off before fitting selector caps. Exercise each input low and high using the approved bias and compatible test signal; observe fresh protocol-v4 status and count transitions. Verify telemetry after closing maintenance. Power cycle again and verify saved ID/inversion/debounce and all three input modes/active conditions.

Copy fixture_measurements.template.json, enter measured voltages and set each check true only after actually observing it. Record live identity/protection plus those results:

```
python host/fixture_record.py --channel can0 --uid UID \
  --measurements measured-unit.json --operator NAME --lot LOT \
  --output unit-UID.json
```

Use `--bridge localhost:5810` for a robot tunnel. An incomplete/failed record exits nonzero. Existing output files are never overwritten. The JSON stores operator/lot/time/UID/settings and a hash of the supplied measurements; keep it with the programming log. It does not independently measure voltages or validate operator assertions.

Design qualification still needs 6/12/22 V input tests, both rails at load, shorts, temperatures, representative sensors/cables, vibration, actual flash power cuts, bus-off/controller disconnect and boot/interrupt timing. See VALIDATION.md. Complete the pilot before calling units production-ready or quoting a qualified manufacturing yield.

Exercise mixed software modes and verify floating detection without false counts using INPUT_MODES.md. Unsettled inputs must not appear as valid robot measurements.

13 Connectors

Source: docs/CONNECTORS.md

References	Exact MPN	Series description	DigiKey
J1-J3	**1824420000**	LSF-SMT 3.50/02/90 3.5SN BK TU	281-2020-ND
J4-J6	**1824430000**	LSF-SMT 3.50/03/90 3.5SN BK TU	281-2021-ND

Manufacturer pages: 2-pole, 3-pole. `/90` gives horizontal wire entry parallel to the PCB. `/135` is the angled version and is not used. Phoenix PTSM is not fitted.

These are through-hole/THR parts despite “SMT” in the family name. Pitch is 3.5 mm, with two solder legs per pole and 7 mm row spacing. PCB drills are 1.1 mm. Housing widths are 7.7 mm / 11.2 mm. Manufacturer CAD is 14.3 mm deep; the footprint conservatively reserves 14.75 mm. Height is 8.5 mm above the mounting surface; nominal pin tips reach 3.5 mm below that surface (about 1.9 mm beyond a 1.6 mm PCB).

Current renders use the official STEP solids with a rigid translation (-2.13,-12.625,-3.5) mm into the footprint origin. They are not approximate connector boxes. Display colors are assigned by the project. Original files, transformed files, URLs, hashes, pin/drill checks and nominal bounds are in vendor_connectors.json. Script `import_connector_models.py` checks all six placed pad patterns and the two vendor models. Do not regenerate models merely to change routing: STEP exporters can alter file headers without changing geometry.

Top terminals face outward at the power/CAN edge; three sensor terminals face outward at the opposite edge. Conservative complete footprint envelope is 52 ×72.1 mm; add room for cable bend, strain relief and button access. Renders do not validate conductor retention, tool clearance in an enclosure, molding tolerances or actual solder fit.

Rev F voltage headers

JP101/JP201/JP301 use Harwin M22-2010305: 2 mm pitch, 0.50 mm square pins, 6 × 2 mm body, nominal 5 mm height above the PCB and 3.4 mm tails. The project’s 2 mm header footprint uses 0.8 mm drills and a 0.25 mm nominal body courtyard margin. Pads 1-3 run left-to-right at Y=51.5 mm; pin-1 positions are X=5, 21 and 37 mm.

The Sullins SPN02SVEN-RC drawing specifies a closed top, 2 mm pitch, 0.50 mm square mating pins and nominal 4 × 2 × 4.92 mm body. Contact height is 2.45 mm from the entry, within the header’s 3.5 mm mating pin length. Nominal mating geometry agrees; physical fit and retention remain untested. Caps are delivered loose and not included in the assembled-board STEP model. Use the finished cap envelope for enclosure access/clearance and allow room to remove them.

The generic local header STEP represents the nominal pin/body geometry, not manufacturer-qualified tolerances. Vendor connector hashes continue to cover the unchanged six terminal models. See wiring for one-cap-per-header configuration and vibration retention qualification.

14 Cost reduction

Source: docs/COST_REDUCTION.md

Rev F implements the approved protected boost, passive sourcing and PCB-area reductions. The existing connectors, removable voltage jumpers, electronic input modes and top-side-only assembly are retained.

Change	Ten-board purchasing saving per board
Protected 5 V → 12 V stage and shared-part price breaks	$6.66 (original target $3.60-5.60)
Passives outside the replaced stage	$1.30 (target $1-2)
Total parts saving against Rev E	About $7.96; about 22.7%
PCB area: 52 × 70 → 52 × 64 mm	8.6% less area; dollar saving requires a quotation

The unrounded baseline is $35.1354 and Rev F is $27.1738; separately rounded displayed prices differ by $7.97. Shared MPN price breaks are attributed to the power-stage line so they are not counted twice. Reproducible comparison.

Boards	Parts purchasing per board	Parts purchasing for the lot
1	$47.80	$47.80
5	$30.93	$154.67
10	$27.17	$271.74
25	$24.98	$624.61

These September 8, 2026 observations include supplier minimums, terminals, headers and three loose caps. PCB fabrication, assembly, attrition, programming/test, freight, tax and duty are extra. First-tier consumed parts total $34.56; buying minimum packs for one board costs $47.80. The cost CSV, per-MPN purchases, supplier observations and BOM expose the arithmetic. This is not a PCBA quotation or allocated stock.

What changed outside the power stage

References	Change	Electrical scope
C101, C201, C301	Same CC0805JRNPO9BN102, sourced from LCSC C62774	1 nF, 50 V, C0G, 0805 unchanged
C7, C23	CC0805KKX7R9BB105 → Samsung CL21B105KBFNNNE, C28323	1 µF, 50 V, X7R, 0805 unchanged
C8, C24	TMK212AB7475KG-T → Samsung CL21B475KAFNNNE, C98195	4.7 µF, 25 V, X7R, 0805 unchanged

The LDO capacitor screen assumes 50% DC-bias retention and further temperature/tolerance/aging allowances, leaving about 1.71 µF against a 1 µF requirement. This is an engineering allowance, not a measured manufacturer curve. Validate effective capacitance/ESR and startup on the prototype. Precision input resistors, crystal loading, MCU, CAN and protection parts outside the replaced stage are retained.

Assembly and delivery

There are 110 top SMT placements across 48 exact MPN types and nine user-soldered through-hole parts (39 joints). The new boost has visible VSSOP leads and removes the former QFN power package, Kelvin shunt and four 0402 capacitors. Fewer placements and simpler inspection may help, but the published Standard setup/stencil/feeder/fixture subtotal remains $123.63 per order before the other charges. A smaller PCB can also stay in the same minimum-price tier.

Partial assembly scenarios use the published JLCPCB fee schedule. Economic eligibility, exact part matching, procurement charges and allocation need confirmation. LCSC stock does not establish JLCPCB Basic/Extended classification or order eligibility. Request the same top-SMT-only scope from the chosen Chinese assembler; the user fits the large terminals and power headers.

No supplier has been contacted and no order or delivered quotation is included. Delivery inside one month cannot be confirmed without the lot quantity, destination, allocated parts, assembly slot and courier/customs allowance. Factory handoff.

15 Tools

Source: TOOLS.md

The saved native KiCad schematic/PCB are authoritative. Circuit data, schematic, BOM and PCB must agree. Historical placement/router experiments have been removed; validation never reroutes the board.

Tested toolchain: KiCad 10.0.6 and its matching pcbnew interpreter, Python 3.12, Arm GNU GCC 13.2.1, GNU make, host GCC/G++, Java 17, Java 25 for the WPILib 2027 adapter, Poppler, InteractiveHtmlBom 2.11.2 and pinned Python packages. Native tool binaries and dependency caches are outside Git. PDF generation uses DejaVu Sans/Mono.

```
# Run from current/, after installing native tools.
uv venv --python python3.12 .venv
uv pip sync --python .venv/bin/python tools-requirements.lock
source tools/env.sh
python tools/fetch_driver_dependencies.py
CANDI_JDK25=/path/to/jdk25 bash scripts/validate_release.sh
```

`tools/env.sh` uses this directory's venv, or the repository venv. It prepends `$CANDI_TOOLS_ROOT/bin` (default `$HOME/.local/share/candi-tools/bin`). Expected wrappers include `kicad-cli`, `kicad-python`, `host-gcc`, `host-g++`, `ibom` and Arm compiler commands. `kicad-python` must match the installed KiCad binary bindings. Java 17 can be selected with `CANDI_JDK`; the 2027 test downloads its pinned dependencies.

The validation script refreshes electrical budgets, loop sensitivity, board constraints, firmware/driver/bridge tests, native ERC/DRC/parity, manufacturing exports and independent Gerber geometry/connectivity checks. It compares against the small frozen Rev E contract in `hardware/design/rev_e_baseline.json`, without depending on an expanded historical checkout.

After validated changes, refresh documents and inspect the results:

```
bash scripts/render_board.sh
python scripts/compare_assembly_cost.py
python scripts/build_documents.py
python scripts/build_handbook.py
python scripts/review_documents.py
python scripts/build_release.py
python scripts/verify_archive.py
```

`build_documents.py` creates the factory packet and Gerber ZIP. `build_release.py` writes a curated source/review archive and manifest. `verify_archive.py` extracts it, verifies all hashes and reruns native checks. The manifest excludes itself and its enclosing ZIP from its own hashes. Review PDF page bounds and rendered pages; automated text bounds cannot detect every overlap.

Optional desktop setup preview: `python scripts/check_gui.py` with a working display, Tkinter and Pillow. It uses simulated data and performs no CAN I/O. `scripts/generate_schematic.py` rebuilds native sheets from `hardware/design/circuit.py`; PCB edits remain a separate CAD operation. `import_connector_models.py` is only for restoring/reviewing exact vendor CAD and transforms, followed by fresh STEP/render exports.

Compiler objects, downloaded Java dependencies, test runners, caches and scratch renderings are ignored. The retained firmware third-party subset supports STM32F042 and GCC; see its provenance. All checks are engineering evidence; physical qualification remains required.

16 Sources and notices

Source: docs/SOURCES_AND_LICENSES.md

KiCad library/model notices are retained under `licenses/`, together with Arm CMSIS, ST device-header and WPILib notices. Firmware third-party directories and InteractiveHtmlBom output retain their notices. Original project work has no selected blanket license. Manufacturer documents/CAD remain their owners' material; retaining a file is not a redistribution-license grant.

Subject	Primary reference
Input fuse interruption / derating	Littelfuse 451/453
Buck and capacitors	TI LMR51420
Regulated 12 V boost	TI TPS61085, exact DGK VSSOP package
Shielded 3.3 µH inductor	Bourns SRP5030TA
Input/output/bias capacitor curve	Samsung CL32B106KBJNNW
Banner supply/residual voltage exceptions	QS18 product manual
Sensor 5 V / boost-input latch-off limiters	TI TPS2553, -1 variant
Comparator/leakage caveats	TI LM339LV
CAN interface / bypass	TI TCAN3413
STM32 pinout, supplies, clock	ST STM32F042
Example industrial output residual levels	Omron E3Z specifications
Terminal geometry and exact MPNs	Weidmüller 1824420000, 1824430000
CAN addressing / broadcast disable	WPILib

Samsung CL31A226KAHNNNE typical characteristic data is retained in `sources/datasheets/samsung_CL31A226_characteristics.json`, inherited from the earlier design's manufacturer retrieval. Typical curves do not establish guaranteed capacitance. DigiKey URLs, exact codes and dated observations are in `bom/catalog_china.json`. Vendor STEP URLs, hashes and transforms are in `hardware/library/3d/vendor_connectors.json`.

Equivalent parts need engineering review; distributor substitution suggestions do not establish electrical/mechanical equivalence. Availability and season rules can change after these observations.

Local source evidence includes the TI TPS61085/TPS2553 datasheets and Samsung CL31A226/CL32B106 characteristic JSON. Typical curves do not establish guaranteed minimum capacitance. U8's footprint and nominal STEP body are from the KiCad 10 Package_SO library. Horizontal connector STEP models are official vendor geometry. The new Samsung CL21B105/CL21B475 substitutions retain nominal ratings; the LDO derating allowance is analytical, not a measured characteristic curve.

Rev F sourcing URLs and tiers are in catalog_china.json. Header/cap geometry links the Harwin and Sullins manufacturer drawings; cost assumptions links the assembly fee and lead-time sources. Stock observations are not factory allocations.

Rev F input bias references: AO3401A, BSS138BK, BAT46W. The local SOT-23 footprint/model comes from installed KiCad libraries under their existing library-license policy. CTRE input-mode API informed feature choices; no CTRE firmware or hardware design is included.